# SC12 - Revision H purchasing BOM

140 fitted parts: 130 factory SMT + 10 user-fitted through-hole parts. Three loose caps extra. Parts purchasing estimate: $28.99/board at 10 boards; PCB, assembly, attrition, freight and tax extra. See cost_estimate.csv. Unbuilt prototype.

| Qty | MPN / supplier part | References | Value / package | Extended USD first tier |
|---|---|---|---|---|
| 1 | 0451003.MRL / C13982 | F1 | 3A FAST / Fuse_Littelfuse-NANO2-451_453 | 0.2556 |
| 4 | 0805W8F1001T5E / C17513 | R4, R110, R210, R310 | 1k / R_0805_2012Metric | 0.0168 |
| 12 | 0805W8F1002T5E / C17414 | R3, R21, R22, R40, R41, R105, R205, R305, R52, R71, R75, R31 | 10k / R_0805_2012Metric | 0.0468 |
| 8 | 0805W8F1003T5E / C149504 | R50, R111, R112, R211, R212, R311, R312, R70 | 100k / R_0805_2012Metric | 0.048 |
| 1 | 0805W8F3300T5E / C17630 | R30 | 330 / R_0805_2012Metric | 0.0054 |
| 3 | 1824420000 / 281-2020-ND | J1, J2, J3 | 2P PUSH 3.5mm / Weidmuller_LSF_90_2P | 4.89 |
| 3 | 1824430000 / 281-2021-ND | J4, J5, J6 | 3P PUSH 3.5mm / Weidmuller_LSF_90_3P | 6.63 |
| 1 | ABM3B-8.000MHZ-B2-T / 535-9720-1-ND | Y1 | 8MHz 18pF / Abracon_ABM3B_5.0x3.2mm | 0.76 |
| 3 | AO3401A / C15127 | Q101, Q201, Q301 | AO3401A / SOT-23 | 0.285 |
| 1 | AO4485 / C51499 | Q1 | AO4485 / SOIC-8_3.9x4.9mm_P1.27mm | 0.2971 |
| 1 | AP2112K-3.3TRG1 / AP2112K-3.3TRG1DICT-ND | U2 | AP2112K-3.3 / SOT-23-5 | 0.28 |
| 2 | B240A-13-F / C92506 | D4, D5 | B240A / D_SMA | 0.192 |
| 3 | BAT46W-E3-18 / C5557780 | D102, D202, D302 | BAT46W / D_SOD-123 | 0.3921 |
| 5 | BSS138BK,215 / C282529 | Q102, Q202, Q302, Q2, Q3 | BSS138BK / SOT-23 | 0.337 |
| 2 | C0805C300J5GACTU / 399-C0805C300J5GACTUCT-ND | C25, C26 | 30pF / C_0805_2012Metric | 0.66 |
| 3 | CC0805JRNPO9BN102 / C62774 | C101, C201, C301 | 1nF / C_0805_2012Metric | 0.0597 |
| 1 | CC0805KRX7R9BB333 / 311-1374-1-ND | C69 | 33nF / C_0805_2012Metric | 0.14 |
| 2 | CL21B103KBANNNC / C1710 | C72, C73 | 10nF / C_0805_2012Metric | 0.0262 |
| 14 | CL21B104KCFNNNE / C28233 | C3, C6, C11, C20, C21, C22, C27, C40, C41, C50, C51, C66, C74, C75 | 100nF / C_0805_2012Metric | 0.5754 |
| 5 | CL21B105KBFNNNE / C28323 | C7, C23, C61, C65, C71 | 1uF / C_0805_2012Metric | 0.2215 |
| 1 | CL21B473KBCNNNC / C53134 | C70 | 47nF / C_0805_2012Metric | 0.0174 |
| 2 | CL21B475KAFNNNE / C98195 | C8, C24 | 4.7uF / C_0805_2012Metric | 0.1222 |
| 8 | CL31A226KAHNNNE / C12891 | C4, C5, C13, C12, C60, C62, C63, C64 | 22uF / C_1206_3216Metric | 1.3672 |

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|---|---|---|---|---|
| 1 | CL32B106KBJNNWE / C3844168 | C2 | 10uF / C_1210_3225Metric | 0.4984 |
| 1 | EEE-FK1H470P / PCE3815CT-ND | C1 | 47uF / CP_Elec_8x6.5 | 0.93 |
| 1 | ESD2CAN24DBZRQ1 / 296-ESD2CAN24DBZRQ1CT-ND | D20 | ESD2CAN24-Q1 / SOT-23 | 0.87 |
| 1 | LM339LVDR / 296-LM339LVDRCT-ND | U7 | LM339LV / SOIC-14_3.9x8.7mm_P1.27mm | 0.42 |
| 1 | LMR51420XDDCR / C5246146 | U1 | LMR51420XDDCR / TSOT-23-6 | 1.0292 |
| 3 | M22-2010305 / 952-2274-ND | JP101, JP201, JP301 | 5V / 12V / PinHeader_1x03_P2.00mm_Vertical | 0.66 |
| 1 | MMSZ12T1G / MMSZ12T1GOSCT-ND | D3 | 12V ZENER / D_SOD-123 | 0.17 |
| 1 | PM254V-12-10P-H85 / C492399 | J8 | 2x5 SERVICE SOCKET 2.54mm / Service_2x05_P2.54mm | 0.1158 |
| 1 | RC0805FR-0711K5L / 311-11.5KCRCT-ND | R61 | 11.5k / R_0805_2012Metric | 0.1 |
| 1 | RC0805FR-0713KL / 311-13.0KCRCT-ND | R63 | 13k / R_0805_2012Metric | 0.1 |
| 3 | RC0805FR-0720KL / 311-20.0KCRCT-ND | R103, R203, R303 | 20k / R_0805_2012Metric | 0.3 |
| 3 | RC0805FR-07220KL / 311-220KCRCT-ND | R101, R201, R301 | 220k / R_0805_2012Metric | 0.33 |
| 1 | RC0805FR-072K61L / C273965 | R62 | 2.61k / R_0805_2012Metric | 0.004 |
| 3 | RC0805FR-072M2L / 311-2.20MCRCT-ND | R104, R204, R304 | 2.2M / R_0805_2012Metric | 0.3 |
| 1 | RC0805FR-0759KL / 311-59.0KCRCT-ND | R10 | 59k / R_0805_2012Metric | 0.1 |
| 1 | RC0805FR-075K23L / 311-5.23KCRCT-ND | R51 | 5.23k / R_0805_2012Metric | 0.1 |
| 1 | RC0805JR-070RL / 311-0.0ARCT-ND | R20 | 0 / R_0805_2012Metric | 0.1 |
| 3 | RC1206FR-0710KL / 311-10.0KFRCT-ND | R109, R209, R309 | 10k / R_1206_3216Metric | 0.33 |
| 1 | RC1206FR-07120RL / 311-120FRCT-ND | R42 | 120 / R_1206_3216Metric | 0.1 |
| 6 | RT0805BRD07100KL / YAG1759CT-ND | R60, R65, R66, R68, R72, R73 | 100k / R_0805_2012Metric | 0.6 |
| 1 | RT0805BRD07102KL / YAG4817CT-ND | R1 | 102k / R_0805_2012Metric | 0.1 |
| 1 | RT0805BRD0712K1L / C865169 | R69 | 12.1k / R_0805_2012Metric | 0.0531 |

| Qty | MPN / supplier part | References | Value / package | Extended USD first tier |
|---|---|---|---|---|
| 1 | RT0805BRD0713K7L / YAG4829CT-ND | R2 | 13.7k / R_0805_2012Metric | 0.1 |
| 2 | RT0805BRD0720KL / C469660 | R67, R74 | 20k / R_0805_2012Metric | 0.1498 |
| 1 | SK6805-EC15 / C2890035 | D10 | SK6805-EC15 / LED_SK6805_EC15 | 0.0956 |
| 1 | SMAJ13A-13-F / C110519 | D6 | SMAJ13A / D_SMA | 0.1246 |
| 3 | SMAJ30A / C148230 | D101, D201, D301 | SMAJ30A / D_SMA | 0.4377 |
| 1 | SMBJ22A-E3/52 / SMBJ22A-E3/52GICT-ND | D2 | SMBJ22A / D_SMB | 0.56 |
| 1 | SN74AHCT1G125DBVR / C7484 | U10 | SN74AHCT1G125 / SOT-23-5 | 0.1009 |
| 1 | SRP5030TA-3R3M / SRP5030TA-3R3MCT-ND | L2 | 3.3uH / L_Bourns_SRP5030T | 1.17 |
| 1 | SRP7028A-100M / SRP7028A-100MCT-ND | L1 | 10uH / L_Bourns_SRP7028A_7.3x6.6mm | 1.1 |
| 1 | STM32F042K6T6 / 497-14647-ND | U5 | STM32F042K6T6 / LQFP-32_7x7mm_P0.8mm | 4.34 |
| 1 | TCAN3413DR / C22433320 | U6 | TCAN3413DR / SOIC-8_3.9x4.9mm_P1.27mm | 1.0227 |
| 1 | TPS2553DBVR-1 / C111738 | U4 | TPS2553DBVR-1 / SOT-23-6 | 0.2935 |
| 1 | TPS259620DDAR / C2155773 | U9 | TPS259620DDAR / TPS2596_DDA8 | 0.8227 |
| 1 | TPS61085DGKR / C113659 | U8 | TPS61085DGKR / VSSOP-8_3x3mm_P0.65mm | 0.6001 |